

Network Security & Scanning Lab Report

ApexPlanet Cybersecurity Internship – Task No 2

1. Introduction

This task focused on understanding practical aspects of network security, reconnaissance, vulnerability scanning, and traffic analysis. The objective was to perform real-time scanning and monitoring of a vulnerable system in a controlled lab environment and analyze the findings using professional cybersecurity tools.

The task helped in developing hands-on skills in network enumeration, vulnerability detection, packet analysis, and basic firewall configuration.

2. Lab Environment Setup

A virtual lab was created using VirtualBox to safely simulate an attack and defense scenario.

- **Attacker Machine:** Kali Linux
- **Target Machine:** Metasploitable2
- **Network Mode:** Host-Only Adapter

The IP address of the target system was identified using the ifconfig command.

Target IP Address: 192.168.56.101

(Insert Screenshot: Metasploitable ifconfig showing IP address)

3. Reconnaissance

3.1 Connectivity Verification

To verify communication between the attacker and target machine, a ping test was performed.

Command used:

```
ping -c 4 192.168.56.101
```

The result showed 0% packet loss, confirming that the target system was reachable from Kali Linux.

(Insert Screenshot: Ping test output)

4. Port and Service Scanning

Network scanning was performed using Nmap to identify open ports and running services.

4.1 TCP SYN Scan

Command:

```
nmap -sS 192.168.56.101
```

The scan revealed multiple open ports such as:

- 21 (FTP)
- 22 (SSH)
- 23 (Telnet)
- 80 (HTTP)
- 3306 (MySQL)

(Insert Screenshot: Nmap TCP scan output)

4.2 Service Version Detection

Command:

```
nmap -sV 192.168.56.101
```

This scan detected service versions, including:

- vsFTPD 2.3.4
- Apache Web Server
- Samba Service
- MySQL Database

This information is critical because outdated services may contain vulnerabilities.

(Insert Screenshot: Nmap -sV output)

5. Vulnerability Assessment

A vulnerability scan was conducted using Nmap's vulnerability scripts.

Command:

```
nmap --script vuln 192.168.56.101
```

The scan identified a critical vulnerability:

- **vsFTPD 2.3.4 Backdoor**
- **CVE-2011-2523**
- Status: Vulnerable

This confirms that the FTP service on the target system contains a known backdoor vulnerability.

(Insert Screenshot: Vulnerability showing "VULNERABLE" and CVE ID)

6. Packet Analysis Using Wireshark

Wireshark was used to capture and analyze network traffic between the attacker and the target system.

6.1 ICMP Packet Capture

Wireshark capture was started on interface eth0, and ICMP filter was applied:

icmp

After running the ping command again, ICMP Echo Request and Echo Reply packets were captured successfully.

This demonstrated how packet-level communication can be monitored in real time.

(Insert Screenshot: ICMP packets in Wireshark)

6.2 FTP Credential Capture

To demonstrate insecure communication, FTP login was attempted while Wireshark was capturing traffic.

Filter applied:

ftp

During login using default credentials:

- Username: msfadmin
- Password: msfadmin

Wireshark successfully captured the USER and PASS commands in plaintext.

This clearly shows that FTP does not encrypt credentials and is insecure.

(Insert Screenshot: USER and PASS visible in Wireshark)

7. Firewall Implementation

To understand basic firewall configuration, nftables was used to create a rule blocking HTTP traffic (Port 80).

Commands used:

```
Sudo nft add table inet myfilter
sudo nft add chain inet myfilter input { type filter hook input priority 0 \; }
sudo nft add rule inet myfilter input tcp dport 80 drop
```

The rule was verified using:

```
sudo nft list ruleset
```

This demonstrates how firewall rules can be implemented to restrict traffic.

(Insert Screenshot: nft ruleset showing tcp dport 80 drop)

8. Results Summary

The following were successfully performed:

- Verified target connectivity
- Identified open ports and services
- Detected vulnerable FTP service
- Captured ICMP traffic
- Captured FTP credentials in plaintext
- Implemented firewall rule using nftables

9. Conclusion

This task provided practical exposure to network security concepts and tools used in real-world penetration testing and network defense. It demonstrated how attackers identify open services and vulnerabilities, and how network traffic can reveal sensitive information if not properly encrypted.

The exercise also highlighted the importance of secure protocols and firewall configurations in protecting systems from potential threats.

Overall, this task strengthened my understanding of reconnaissance, scanning, vulnerability assessment, packet analysis, and firewall implementation in a controlled cybersecurity lab environment.